Circular Packaging Protocols in Consumer Supply Chains.

Transitioning fast-moving consumer goods to circular packaging frameworks requires substituting virgin petroleum polymers with post-consumer recycled polyethylene terephthalate and industrially compostable polyhydroxyalkanoates. Achieving mechanical recyclability without sacrificing barrier properties against moisture and oxygen ingress represents a critical chemical engineering challenge. Successful circular supply loops require closed-loop reverse collection incentives, automated near-infrared optical flake sorting, and decontamination extrusion technologies to meet food-grade contact safety regulations.